JERFIN K

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PROFESSIONAL SUMMARY

Electronics and Communication Engineer building robotics and computer-vision software in Python, C/C++, and Java across autonomous drone, surface (ASV), and underwater (AUV) platforms. Hands-on experience in multi-sensor integration, embedded control, PID control, and real-time telemetry systems, gained at Xera Robotics and the National Institute of Ocean Technology (NIOT), Chennai. Selected participant, e-Yantra Robotics Competition, IIT Bombay. Seeking an Autonomous Systems, Embedded Software, or Robotics Software Engineering role.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, Dart
Robotics & Simulation: ROS2, Gazebo, Turtlesim, PID Control, Autonomous Navigation, Path Planning, Pixhawk, MAVLink
Computer Vision & AI: OpenCV, YOLO, MediaPipe, Object Detection, Face Recognition
Embedded & Hardware: ESP32, Arduino, Raspberry Pi 4, Jetson Orin NX/Nano, Sensor Integration, Motor Drivers
Protocols & Networking: UART, I2C, SPI, RS232, TCP/IP, UDP, Bluetooth
Core CS & Practices: Data Structures & Algorithms, OOP, Debugging, Unit Testing, SDLC, Agile, Technical Documentation
Tools & Platforms: Git, GitHub, Linux, QtPy, Flutter, Firebase, SQL

PROFESSIONAL EXPERIENCE

Software Engineer | Xera Robotics, Chennai **Jan 2026 - Present**

- Built real-time computer vision and monitoring features (Python, OpenCV) for Vanachitram, an AI-powered drone wildlife-surveillance system deployed in Tirumala Tirupati Devasthanams forest regions.
- Integrated Pixhawk flight-controller and drone communication systems to enable autonomous flight operations.
- Validated AI object-detection models through structured, repeatable test cycles, improving detection reliability for field deployment.

Junior Firmware Engineer | National Institute of Ocean Technology (NIOT), Chennai (via Xera Robotics) **Feb 2026 - Jul 2026**

- Developed Autonomous Surface Vehicle (ASV) and Autonomous Underwater Vehicle (AUV) systems integrating live telemetry from 5+ sensor types (GPS, AHRS, sonar, pressure) for real-time data acquisition.
- Integrated u-blox GPS, Xsens AHRS, Ping2 altimeter, BlueRobotics Bar30, Ping360 sonar, LoRa/Wi-Fi, and Moxa modules across ASV/AUV platforms.
- Built a QtPy-based ground-station application with mission-planning capability, delivering end-to-end features from telemetry ingestion through UI visualization and field testing.
- Documented system architecture, sensor wiring, and data-flow diagrams to support integration across mechanical, software, and electronics teams.
- Led debugging and performance testing across 3 autonomous vehicle platforms, resolving reliability issues ahead of field deployment.

Robotics & IoT Intern | Sristhi Innovative Robotics, Thiruvananthapuram **Jul 2025 - Aug 2025**

- Built obstacle-avoiding, line-following, and autonomous delivery robots using Arduino sensor interfacing with live monitoring.
- Developed a fire-alert system with SMS notification and Firebase real-time logging, and a Raspberry Pi/OpenCV face-detection attendance system.
- Delivered embedded and cloud-integrated applications using ESP32, Firebase, Python, and OpenCV.

PROJECTS

- **JerGroundStation (QtPy, Python, MAVLink):** Custom ground-control-station software for real-time UAV/robot telemetry visualization and camera streaming; implemented MAVLink two-way vehicle communication.
- **AI-Based Autonomous Drone Sensor Integration (Pixhawk, UART, I2C, Python):** Integrated GPS, AHRS, camera, and environmental sensors with the flight controller; performed sensor calibration and telemetry monitoring for real-time flight data processing.
- **Drone Simulation (ROS2, Gazebo, Turtlesim):** Simulated drone flight dynamics for 3D visualization and motion-control testing; validated topic publishing and command-based movement.

EDUCATION

Bachelor of Engineering, Electronics and Communication Engineering **Expected Nov 2026**
University College of Engineering, BIT Campus, Tiruchirapalli

ACHIEVEMENTS AND LEADERSHIP

- Selected, e-Yantra Robotics Competition, IIT Bombay (Krishi Drone theme) - developed PID-based drone control algorithms in ROS2/Gazebo with sensor-based autonomous navigation.
- Led Team ALPHA_PI to build an offline home-automation prototype on Raspberry Pi within a 24-hour national hackathon (SPRINTATHON'25).
- Built and tested an autonomous robot for PacBot, Kurukshetra '24, Anna University Chennai.
- National Service Scheme (NSS) - Overall In-Charge; led campus and community outreach programs.